# Frequency-Lattice Inversion of Multi-Prism Risley Systems: Exact Parameter Recovery with Information-Theoretic Certificates

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(Dated: July 21, 2026)

We solve the full parameter-recovery inverse problem for multi-prism Risley beam-steering systems: given one observed scan pattern and nothing else, recover the complete physical specification—rotation speeds, wedge angles, glass indices, system geometry, and beam source ($4P+6$ parameters for $P$ prisms; 18 for the reference case $P$=3). The key structural fact is that the pattern's analytic signal $z = x + iy$ is quasi-periodic on the rank-$P$ integer lattice generated by the *signed* prism speeds, and every quantity the inverse problem needs is a property of that lattice model: the generators are the speeds (sign included), the phases of the fundamental lines are the wedge orientations exactly, and their amplitudes determine the wedge magnitudes. The resulting constructive inversion—matrix-pencil line estimation, CLEAN-style line growth, integer-coverage basis identification, variable projection, and one trust-region completion—contains no enumeration and no training: no sign combinations, no frequency-triple search, no angle grids, no learned components. Every answer is *certified*: per-parameter error bounds derived from the exact-model Jacobian, or a Fisher-information impossibility statement naming the deficit—pair resolvability, prism detectability, lattice degeneracy, mask budget, front-end capacity—together with the observation time that cures it. We further derive a rigorous scaling theory in the prism count: per-instance cost is polynomial in $P$; pairwise speed crowding demands observation time $\Theta(P^2)$, and accidental lattice relations demand time polynomial of degree equal to the working relation order, yielding a computable feasibility frontier. Every statistical claim is subjected to a pre-registered large-scale computational campaign—order $10^6$ random configurations under an adaptive observation protocol and order $10^7$ ground-truth certificate checks (Sec. VII); pending values are marked [·] throughout. This is a simulation study; experimental validation is planned.

## I. INTRODUCTION

### A. Background

Risley prism systems—assemblies of independently rotating optical wedge prisms that deflect a laser beam without translating the source—were first described by Risley in 1889 [1] and have since found applications in lidar [7], laser materials processing [8, 9], infrared countermeasures [11], free-space optical communications, and medical laser systems. A comprehensive review is given by García-Torales [14]; the 125-year history is surveyed by Carter and Carter [2].

A physical wedge prism is a solid piece of glass with two non-parallel planar faces. A beam entering through the first (entry) face refracts by Snell's law, propagates through the glass, and refracts again on exit. Each prism thus contributes *two* refractive interfaces rotating in lockstep; a system of $P$ prisms has $2P$ interfaces. The *forward* problem—predicting the scan pattern from known parameters—is well understood, both paraxially [4, 5, 12] and by exact ray tracing [3, 6, 13].

### B. The inverse problem

The *parameter-recovery* inverse problem has received far less attention, and what little exists solves a fundamentally different problem. What the literature calls the Risley "inverse problem" almost universally denotes *beam steering*: given *known* hardware, compute the prism orientations that place the beam on a target—solved in closed form for two prisms in the paraxial limit [5], by look-up table [10], and most recently by neural networks [18, 19]. This is a pointing problem, not an identification problem: the hardware is an input, not an output.

A smaller line of work does estimate hardware, but only as bounded corrections to a *known design*. Li *et al.* [15] identify six error parameters of a two-prism pointer (wedge angles, indices, installation angles) as small deviations around a known nominal, from controlled pointing measurements on an experimental platform, by genetic search, improving static pointing from 33 to under 1 arcsec; pointing-error compensation on an assembled unit [16] and target-free self-calibration of prism main-section orientation from the Fourier spectrum [17] likewise recover partial, near-nominal quantities for two prisms—never the wedge magnitudes, indices, and rotation dynamics together, and never blind.

Our problem is different in kind: nothing is nominal and nothing is controlled. A time series of workpiece positions is passively observed, and the *complete* physical specification—rotation speeds (dynamics, absent from every calibration setting above), both wedge-angle components per prism, glass indices, layout distances, and the beam source—is recovered blind, to near machine precision, with certified error bounds. For this full system-identification problem—blind, from a single passively observed pattern, for $P \geq 2$ prisms—no prior solution has been reported.

The difficulty is threefold: the map is nonlinear (iterated Snell refraction at $2P$ interfaces); the parameter space has $4P + 6$ dimensions (18 for $P$=3); and the problem is severely ill-conditioned—the Jacobian condition number $\kappa \approx 5 \times 10^5$ means geometry parameters are roughly five orders of magnitude less sensitive than speeds (Sec. III C). In our own earlier work the missing piece was never the local optimizer—a trust-region method on the exact model reaches near machine precision from any initialization in the correct basin—but the *global* stage that finds that basin. Successive attempts based on FFT peak picking required layers of brute force to survive their own failure modes: an enumeration of candidate frequency triples (the top spectral peaks are often harmonics or intermodulation lines rather than fundamentals), a $2^P$ enumeration of speed signs (real-valued spectra cannot see rotation direction), and a grid over wedge-angle space (bin-limited spectral phases are corrupted by $\mathcal{O}(90°)$ discretization error, so angles could not be read from the spectrum). This paper removes all three layers, not by better search, but by identifying the algebraic structure that makes search unnecessary. Figure 1 shows the end result on one representative random instance: a dense multi-lobed scan pattern, the lattice-valued line spectrum the method reads, and the reproduction from the recovered 18-parameter vector—indistinguishable from the observation at MSE $2 \times 10^{-26}$, obtained in 2.9 s.

### C. Contributions

1. **Structure theory** (Sec. III). Away from an explicitly characterized total-internal-reflection (TIR) set, the analytic signal of the scan pattern is quasi-periodic on the integer lattice generated by the *signed* speeds (Theorem 1); the fundamental line of prism $i$ sits at signed frequency $N_i$ and dominates its conjugate (Prop. 3); its phase equals the wedge orientation $\alpha_{y,i}$ exactly up to an exact discrete symmetry (Props. 1–4); and its amplitude is an odd cubic in $\tan \alpha_{x,i}$ (Prop. 5). A finer-lattice lemma (Lemma 2) explains why residual-guided model selection provably cannot identify the generators, forcing the arithmetic selection we use.

2. **An enumeration-free constructive inversion** (Sec. IV): matrix pencil [20, 21] line estimation, CLEAN-style [24] line growth at lattice order one, integer-coverage basis identification, variable-projection [25] lattice refinement, spectral angle readout, and a single trust-region completion [30] with a deterministic, MSE-verified candidate ladder.

3. **Certificates** (Sec. V). Every instance returns either all $4P+6$ parameters with per-parameter error bounds from the exact-model Jacobian, or a quantitative impossibility statement from the Fisher information of the fitted lattice model, with the observation time $T$ that would cure it. Bound coverage and tightness are measured against ground truth at campaign scale (Sec. VII, E2), and the prescriptions are validated by bisection (E4).

4. **Scaling theory for arbitrary prism count** (Sec. VI). Exact and asymptotic laws for the $4P+6$-dimensional problem: per-instance cost polynomial in $P$; an exact pair-crowding law forcing observation time $\Theta(P^2)$; anti-concentration and pigeonhole bounds for accidental lattice relations giving a degree-$K$ polynomial demand at working relation order $K$; a capacity threshold; and the resulting feasibility frontier (Theorem 2), tested head-on by the campaign's $P \times T$ experiment (E5).

5. **Tested assumptions at scale.** Every mathematical claim used by the method is verified by a named experiment (A1–A8, Table I) run over the campaign ensemble, including a direct measurement of the trust-region basin that retires the folklore of "narrow basins" for this problem.

## II. PHYSICAL PRISM MODEL

### A. System geometry

We consider $P$ rotating wedge prisms along the optical axis $z$ (Fig. 2). Each prism $i$ has refractive index $n_{g,i}$ and two planar faces: a flat **entry face** (interface $2i-1$, perpendicular to $z$), and an **exit face** (interface $2i$) tilted by the wedge angle pair $(\alpha_{x,i}, \alpha_{y,i})$, where $\alpha_x$ is the tilt magnitude and $\alpha_y$ the azimuthal orientation offset. Both faces rotate rigidly at speed $N_i$ (Hz),

$$\gamma_i(t) = 360\, N_i\, t \pmod{360°}, \tag{1}$$

and the refractive indices alternate along the path, $n_0 = 1$, $n_1 = n_{g,1}$, $n_2 = 1, \ldots$, so every interface has a nontrivial Snell ratio. The beam originates at $(r_x, r_y, 0)$ with angles $(\theta_x^{(0)}, \theta_y^{(0)})$; inter-element distances are source-to-first-face $d_0$, prism thickness $\tau_p$, inter-prism gap $\delta$, and last-face-to-workpiece $d_W$. Of these, $d_0$ and $\tau_p$ are held fixed at known nominal values and are not recovered; the unknown vector for $P$=3 is

$$\begin{aligned}\boldsymbol{\theta} = [\, & N_1, N_2, N_3,\ \alpha_{x,1..3},\ \alpha_{y,1..3},\ n_{g,1..3}, \\ & d_W,\ \delta,\ \theta_x^{(0)}, \theta_y^{(0)},\ r_x, r_y \,] \in \mathbb{R}^{18}.\end{aligned} \tag{2}$$

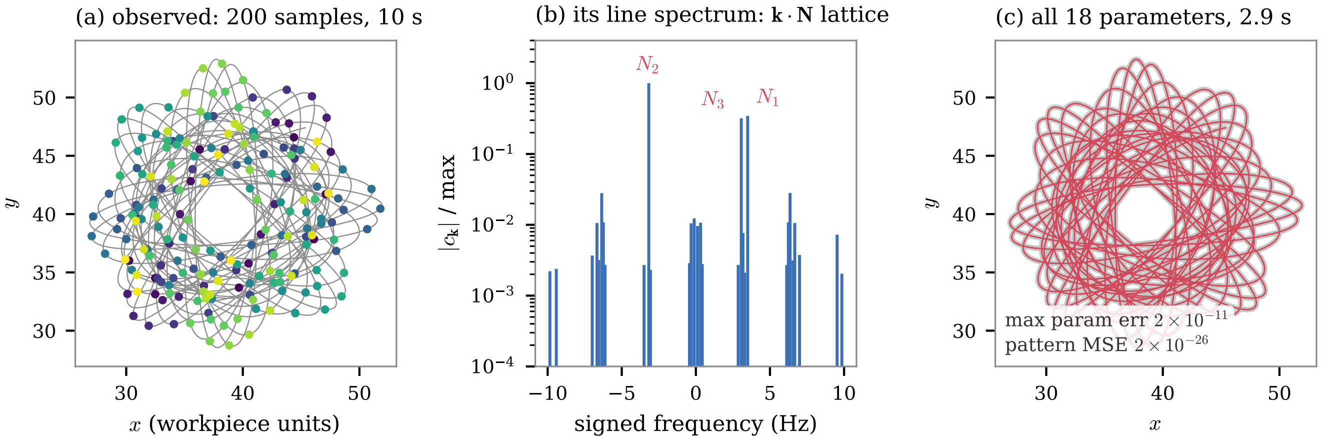


FIG. 1. The method in one instance (a representative random configuration; all quantities as measured). (a) The observed data: 200 samples over 10 s of the workpiece position (dots, colored by time; gray curve is the underlying continuous trajectory). (b) What the method reads: the analytic signal's line spectrum lives on the integer lattice $\{\boldsymbol{k}\cdot\boldsymbol{N}\}$ generated by the three *signed* rotation speeds (labeled fundamentals; the remaining lines are their harmonics, intermodulation products, and sampling aliases, all explained by the same three generators). (c) The pattern regenerated from the recovered parameter vector $\hat{\boldsymbol{\theta}}$ (red) overlaid on the observation (gray): all 18 parameters in 2.9 s, maximum parameter error $2\times10^{-11}$, pattern MSE $2\times10^{-26}$.

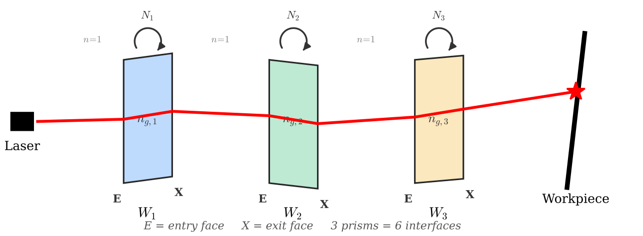


FIG. 2. Three-prism Risley system. Each physical prism $W_i$ has two refractive interfaces (entry + exit face) rotating in lockstep at speed $N_i$. The beam refracts at each air–glass and glass–air transition.

### B. Refraction and rotation

At interface $k$ with effective tilt $\varphi_k$ the direction is updated by vector Snell's law in per-axis reduced form (the $x$ and $y$ projections are traced independently; cross-coupling is $\mathcal{O}(\alpha_x\alpha_y)$, $\sim$7% relative at $\alpha=15°$, and exact for the self-consistent inversion used here): with $\boldsymbol{n}=[\tan\varphi,0,-1]^\top$, $\boldsymbol{s}=[\tan\theta,0,1]^\top$, $\eta=n_{k-1}/n_k$,

$$\boldsymbol{s}_\mathrm{f}=\eta\,(\hat{\boldsymbol{N}}\times(-\hat{\boldsymbol{N}}\times\hat{\boldsymbol{s}}_\mathrm{i}))-\hat{\boldsymbol{N}}\sqrt{1-\eta^2\,|\hat{\boldsymbol{N}}\times\hat{\boldsymbol{s}}_\mathrm{i}|^2}\,. \quad (3)$$

Rotation enters only through the exit-face normal

$$\boldsymbol{n}_i=\begin{bmatrix}\cos(\gamma_i+\alpha_{y,i})\,\tan\alpha_{x,i}\\ \sin(\gamma_i+\alpha_{y,i})\,\tan\alpha_{x,i}\\ -1\end{bmatrix}. \quad (4)$$

Equation (4) is the algebraic seed of the whole method: $\alpha_{y,i}$ appears *only* as a phase offset added to $\gamma_i$, and $\alpha_{x,i}$ *only* through the tilt magnitude $\tan\alpha_{x,i}$. Two exact consequences follow.

**Proposition 1** (Exact flip symmetry)**.** *The forward map is invariant under* $(\alpha_{x,i},\alpha_{y,i})\mapsto(-\alpha_{x,i},\alpha_{y,i}+180°)$ *for any prism, at all orders.*

*Proof.* $\cos(\gamma+\alpha_y+180°)\tan(-\alpha_x)=\cos(\gamma+\alpha_y)\tan\alpha_x$ and likewise for the sine component, so the normal (4) is pointwise unchanged; the trace depends on the parameters only through $\boldsymbol{n}_i(t)$. □

*Test A1* verifies the invariance to floating-point precision over the campaign ensemble: $\max|\Delta F|$ = **[A1]** (Table I).

Because the admissible box $|\alpha_y|\le 18°$ contains exactly one representative of each symmetry class, the flip is a reparameterization, not a non-uniqueness; it does, however, control the branch logic of the spectral angle readout (Sec. IV E).

**Proposition 2** (TIR set)**.** *The trace is* $C^\infty$ *in* $(\boldsymbol{\theta},t)$ *except on the set where* $1-\eta^2|\hat{\boldsymbol{N}}\times\hat{\boldsymbol{s}}_\mathrm{i}|^2=0$ *at some interface (total internal reflection at a glass–air exit face), where the radicand in* (3) *is clamped and the pattern acquires* $C^0$ *kinks and large excursions.*

The TIR set is not exotic: a random draw from the admissible box crosses it during a 10-s scan with probability [$p_\mathrm{TIR}$, **E1**], producing pattern jumps of hundreds of spatial units over a handful of samples. An earlier version of this work asserted global smoothness; the assertion is false as stated, and the inversion must (and does, Sec. IV A) handle the TIR set explicitly.

## III. STRUCTURE THEORY

Throughout, $\boldsymbol{p}(t) = (x(t), y(t))$ is the workpiece trace sampled uniformly at $f_s = 1/\Delta t$ over $[0, T)$, and

$$z(t) = x(t) + i\,y(t) \tag{5}$$

is its analytic combination. Working with $z$ rather than $x, y$ separately is essential: a complex signal has no conjugate symmetry, so positive and negative frequencies are distinct observables.

### A. The frequency lattice

**Theorem 1** (Lattice support). *Fix a parameter vector $\boldsymbol{\theta}$ and let $\Phi : \mathbb{T}^P \to \mathbb{R}^2$ be the workpiece map evaluated at rotor angles $\boldsymbol{\gamma} \in \mathbb{T}^P$, so that $\boldsymbol{p}(t) = \Phi\big(2\pi N_1 t + \alpha_{y,1}, \ldots, 2\pi N_P t + \alpha_{y,P}\big)$. Assume*

- *(H1) at every $\boldsymbol{\gamma} \in \mathbb{T}^P$ and every interface the Snell radicand in (3) is $\geq \rho_0 > 0$ (no TIR anywhere on the rotor torus);*
- *(H2) every ray–face intersection on $\mathbb{T}^P$ is transversal: the intersection denominators are $\geq \rho_1 > 0$.*

*Then there exist $C < \infty$ and $\eta > 0$, depending only on $(\rho_0, \rho_1)$ and the parameter box, such that*

$$z(t) \;=\; \sum_{\boldsymbol{k} \in \mathbb{Z}^P} c_{\boldsymbol{k}}\, e^{2\pi i\,(\boldsymbol{k}\cdot\boldsymbol{N})\,t}, \qquad |c_{\boldsymbol{k}}| \;\leq\; C\, e^{-\eta\,|\boldsymbol{k}|_1}, \tag{6}$$

*with the series absolutely and uniformly convergent.*

*Proof.* Each interface normal (4) depends on $\gamma_i$ only through $\cos\gamma_i$ and $\sin\gamma_i$, which are restrictions to the unit polytorus of the functions $(u_i + u_i^{-1})/2$ and $(u_i - u_i^{-1})/2i$, holomorphic on $(\mathbb{C}^*)^P$ in $u_i = e^{i\gamma_i}$. The trace is obtained from the normals by a *finite* composition of field operations, square roots [Eq. (3)], and two-line intersections (rational). Extend this composition step by step to the polyannulus $A_r = \{r^{-1} \leq |u_i| \leq r\}^P$: at each step, the functions already constructed are holomorphic on $A_r$ and continuous up to the boundary for $r$ close enough to 1; by (H1) each radicand is $\geq \rho_0$ on the compact torus, so by continuity and compactness it stays in the slit plane $\mathbb{C} \setminus (-\infty, 0]$ on $A_r$ for $r$ close to 1, where the principal square root is holomorphic; by (H2) each denominator stays nonzero on $A_r$ likewise. Since the composition has finitely many steps ($2P$ interfaces), a single $r > 1$ works for all of them, and $\Phi_1 + i\Phi_2$ extends holomorphically to $A_r$. For any $\boldsymbol{k}$, apply Cauchy's formula on the polycircle $|u_i| = r^{\operatorname{sign} k_i}$ (radius 1 where $k_i = 0$):

$$\begin{aligned} |c_{\boldsymbol{k}}| &= \Big|\frac{1}{(2\pi i)^P} \oint\cdots\oint \frac{(\Phi_1 + i\Phi_2)(\boldsymbol{u})\,d\boldsymbol{u}}{u_1^{k_1+1}\cdots u_P^{k_P+1}}\Big| \\ &\leq\; \|\Phi_1 + i\Phi_2\|_{A_r}\, r^{-|\boldsymbol{k}|_1}, \end{aligned}$$

which is the stated decay with $\eta = \log r$ and $C = \|\Phi_1 + i\Phi_2\|_{A_r} < \infty$ by compactness. Absolute uniform convergence of the multiple Fourier series follows by summing the geometric bound. Evaluating along the orbit $\gamma_i(t) = 2\pi N_i t + \alpha_{y,i}$ turns the monomial $\boldsymbol{u}^{\boldsymbol{k}}$ into $e^{i\boldsymbol{k}\cdot\boldsymbol{\alpha}_y}\, e^{2\pi i(\boldsymbol{k}\cdot\boldsymbol{N})t}$; absorbing the constant phase into $c_{\boldsymbol{k}}$ gives (6). □

Configurations violating (H1) on part of the torus are exactly the TIR draws of Prop. 2: there the expansion holds on the sub-orbit where the radicands stay positive, and the algorithm masks the complement (Sec. IV A). *Test A4* measures the practical truncation: projecting the exact trace onto $|\boldsymbol{k}|_1 \leq 4$ leaves median relative residual **[A4]** over the campaign ensemble, exceeding $10^{-2}$ essentially only on TIR draws (fraction **[A4b]**).

Three refinements of Theorem 1 carry the inversion.

**Lemma 1** (Paraxial cascade gains). *Linearize the per-axis forward map at zero wedge angles (all $\alpha_{x,j} = 0$), with glass, geometry, and beam fixed. Let $K_{x,i}$ [resp. $K_{y,i}$] denote the derivative of the workpiece $x$ [resp. $y$] coordinate with respect to an $x$-slope [resp. $y$-slope] perturbation injected at the exit face of prism $i$. Then*

$$\begin{aligned} K_{\bullet,i} &= \sum_{s \in \mathcal{S}_i} \ell_s \sec^2\theta_s \prod_{m \in \mathcal{M}_{i,s}} D_m, \\ D_m &= \frac{n_{m^-}\cos\theta_{m^-}}{n_{m^+}\cos\theta_{m^+}} \;>\; 0, \end{aligned} \tag{7}$$

*where $\mathcal{S}_i$ is the set of ray segments downstream of prism $i$, $\mathcal{M}_{i,s}$ the set of interfaces crossed between prism $i$ and segment $s$, and $\theta$ the unperturbed per-axis ray angles. In particular $K_{x,i}, K_{y,i} > 0$; they differ only through the $\theta$-dependence induced by the beam angles.*

*Proof.* At zero wedge every face is perpendicular to $z$, so the per-axis Snell law at interface $m$ is $n_{m^-}\sin\theta_{m^-} = n_{m^+}\sin\theta_{m^+}$; away from TIR both cosines are positive, and implicit differentiation gives $d\theta_{m^+}/d\theta_{m^-} = D_m > 0$. A straight segment of length $\ell_s$ maps $(x, \theta) \mapsto (x + \ell_s\tan\theta, \theta)$, with $\partial x_{\text{out}}/\partial\theta = \ell_s\sec^2\theta$ and $\partial x_{\text{out}}/\partial x = 1$; a zero-wedge interface leaves $x$ continuous. The chain rule along the downstream composition yields exactly (7): each summand is the sensitivity path that propagates as an angle up to segment $s$ and as a position thereafter, and every factor is strictly positive. □

**Proposition 3** (Signed fundamentals and the conjugate leak). *Write $\tau_i = \tan\alpha_{x,i}$ and $\tilde{\alpha}_{y,i} = \alpha_{y,i} + 180°[\tau_i < 0]$. As $\max_j |\tau_j| \to 0$,*

$$z(t) = z_0 + \sum_i \big(A_i\, e^{i\varphi_i(t)} + B_i\, e^{-i\varphi_i(t)}\big) + \mathcal{O}(\tau^2), \tag{8}$$

*with $\varphi_i(t) = 2\pi N_i t + \tilde{\alpha}_{y,i}$, $A_i = \frac{1}{2}(K_{x,i} + K_{y,i})\,|\tau_i|$, $B_i = \frac{1}{2}(K_{x,i} - K_{y,i})\,|\tau_i|$, and the gains of Lemma 1. Since both gains are strictly positive, $|B_i| < A_i$ whenever $\tau_i \neq 0$: the fundamental line at signed frequency $N_i$ strictly dominates its conjugate at $-N_i$.*

*Proof.* By the construction in the proof of Theorem 1, $z$ is jointly analytic in $(t, \boldsymbol{\tau})$ near $\boldsymbol{\tau} = \mathbf{0}$; Taylor-expand to first order. At $\boldsymbol{\tau} = \mathbf{0}$ all faces are flat and the trace is the constant $z_0$ (the static ray). The first-order term in $\tau_i$ is computed from Eq. (4): the transverse part of the exit-face normal is $\tau_i(\cos\varphi_i, \sin\varphi_i)$ with $\varphi_i = \gamma_i(t) + \alpha_{y,i}$, and differentiating the Snell update at a flat interface shows the exit-slope perturbation is $\lambda_i \tau_i(\cos\varphi_i, \sin\varphi_i)$ per axis with a common factor $\lambda_i > 0$ (the refraction contrast; positive away from TIR). Lemma 1 propagates this to the workpiece: $\delta x = K_{x,i}\lambda_i\tau_i \cos\varphi_i$, $\delta y = K_{y,i}\lambda_i\tau_i \sin\varphi_i$ (absorb $\lambda_i$ into the gains). The identity $K_x \cos\varphi + iK_y \sin\varphi = \frac{1}{2}(K_x+K_y)e^{i\varphi} + \frac{1}{2}(K_x-K_y)e^{-i\varphi}$ together with $\tau e^{i\varphi} = |\tau| e^{i(\varphi+180^\circ[\tau<0])}$ gives the displayed form, and $K_x, K_y > 0$ implies $|K_x - K_y| < K_x + K_y$, i.e. $|B_i| < A_i$. □

The sign of every speed is therefore an observable of the complex spectrum—no sign enumeration is ever needed—and the ratio $|B_i/A_i|$ measures the $x/y$ gain asymmetry induced by the beam angles. *Test A2*: median leak ratio **[A2]**; the inequality holds in all but a fraction **[A2b]** of prism instances (the rare violations, strongly non-paraxial cases, are caught by the verified sign test of Sec. IV D).

**Proposition 4** (Phase equivariance and readout)**.** *(i)* Exact: *for every* $\boldsymbol{k}$, $c_{\boldsymbol{k}}(\boldsymbol{\alpha}_y) = c_{\boldsymbol{k}}(\mathbf{0})\, e^{i\boldsymbol{k}\cdot\boldsymbol{\alpha}_y}$*; in particular* $\arg c_{\boldsymbol{e}_i} = \alpha_{y,i} + \arg c_{\boldsymbol{e}_i}(\mathbf{0})$*. (ii)* Paraxial branch: $\arg c_{\boldsymbol{e}_i}(\mathbf{0}) = 180^\circ[\alpha_{x,i} < 0] + \mathcal{O}(\tau)$*, hence* $\arg c_{\boldsymbol{e}_i} = \tilde{\alpha}_{y,i} + \mathcal{O}(\tau)$*.*

*Proof.* (i) $\alpha_{y,i}$ enters the forward map only through the sum $\gamma_i + \alpha_{y,i}$ [Eq. (4)], so the torus map of Theorem 1 satisfies $\Phi^{(\boldsymbol{\alpha}_y)}(\boldsymbol{\gamma}) = \Phi^{(\mathbf{0})}(\boldsymbol{\gamma} + \boldsymbol{\alpha}_y)$, and a translation of the torus multiplies the $\boldsymbol{k}$-th Fourier coefficient by $e^{i\boldsymbol{k}\cdot\boldsymbol{\alpha}_y}$. (ii) By Prop. 3 at $\boldsymbol{\alpha}_y = \mathbf{0}$, $c_{\boldsymbol{e}_i}(\mathbf{0}) = A_i\, e^{i\,180^\circ[\tau_i<0]} + \mathcal{O}(\tau^2)$ with $A_i > 0$; since $|A_i| = \Theta(|\tau_i|)$, the relative size of the remainder is $\mathcal{O}(\tau)$, and arg is Lipschitz on $\{w : |w - A| \le \frac{1}{2}A\}$, giving the stated phase error. □

*Test A3*: median readout error **[A3]** (90th percentile **[A3b]**) over the campaign ensemble. The wedge orientations are read off the spectrum; combined with the wide $\alpha_y$-basin of the completion stage this eliminates any search over orientations.

**Proposition 5** (Amplitude law)**.** *Fix all parameters except* $\tau_i = \tan\alpha_{x,i}$ *and write* $c(\tau_i) = c_{\boldsymbol{e}_i}$*. Under (H1)–(H2), c is analytic and odd on a neighborhood of the admissible tilt interval, and there are constants* $a_i > 0$*,* $b_i$ *(functions of the remaining parameters) with*

$$|c(\tau_i)| \;=\; a_i|\tau_i| + b_i|\tau_i|^3 + \mathcal{O}(|\tau_i|^5). \tag{9}$$

*Proof.* By Prop. 4(i), $|c|$ is independent of $\alpha_{y,i}$, so take $\alpha_{y,i} = 0$. Analyticity in $\tau_i$ is part of the holomorphic-extension argument of Theorem 1 (the parameter enters the same finite composition). Oddness: the flip symmetry (Prop. 1) gives $c_{\boldsymbol{e}_i}(-\tau_i;\, \alpha_{y,i}{=}180^\circ) = c_{\boldsymbol{e}_i}(\tau_i;\, \alpha_{y,i}{=}0)$, while equivariance gives $c_{\boldsymbol{e}_i}(-\tau_i;\, 180^\circ) = -\,c_{\boldsymbol{e}_i}(-\tau_i;\, 0)$; hence $c(-\tau) = -c(\tau)$. An odd analytic function factors as $c(\tau) = \tau\, g(\tau^2)$ with $g$ analytic, and $g(0) = \frac{1}{2}(K_{x,i} + K_{y,i}) > 0$ by Prop. 3. Therefore $|c(\tau)| = |\tau|\sqrt{G(\tau^2)}$ with $G = |g|^2$ analytic and $G(0) > 0$, so $\sqrt{G}$ is analytic near 0; expanding it to first order in $\tau^2$ gives the display with $a_i = |g(0)| > 0$. □

*Test A6*: calibrating $(a_i, b_i)$ by two forward evaluations per prism and inverting the cubic recovers $|\alpha_{x,i}|$ with median error **[A6]** when the calibration uses the true geometry and **[A6b]** (90th percentile **[A6c]**) when it uses only nominal mid-box geometry—inside the measured completion basin (Sec. IV F) in either case.

### B. Why generator identification must be arithmetic

The speeds are the *generators* of the observed line set, and one might hope to select them by fitting candidate generator tuples and comparing residuals. This fails structurally.

**Lemma 2** (Finer-lattice lemma)**.** *Let* $\mathcal{L}(\boldsymbol{g}) = \{\boldsymbol{k}\cdot\boldsymbol{g} : |\boldsymbol{k}|_1 \le B\}$ *and let* $\boldsymbol{g}'$ *be any tuple with* $\mathcal{L}(\boldsymbol{g}) \cap \mathcal{F} \subseteq \mathcal{L}(\boldsymbol{g}')$ *on the observed line set* $\mathcal{F}$*. Then the least-squares residual of the* $\boldsymbol{g}'$*-lattice model is no larger than that of* $\boldsymbol{g}$*.*

*Proof.* The $\boldsymbol{g}'$ design spans a superset of the regressors on $\mathcal{F}$; least squares is monotone under design enlargement. □

A compromise or subharmonic tuple whose lattice happens to cover the strong lines (e.g. $g = 1.096$ absorbing a line at $-3.289$ as $\boldsymbol{k} = -3$ and one at $-1.846$ as $(1, -2)$, an actual development failure of a residual-guided predecessor) therefore always ties or beats the true generators on residual. Selection must instead reward *parsimony*—amplitude mass concentrated at low lattice order—which is an arithmetic property of the integer representation, not a fit property. This is the same reason integer-relation methods (LLL [26]) exist: shortness of the representation, not goodness of fit, identifies a basis. Our coverage score (Sec. IV C) implements exactly this.

### C. Identifiability

Local identifiability at the completion stage is a numerical theorem: the $400 \times 18$ Jacobian of the sampled forward map has full rank 18 at every tested solution, with condition number $\kappa \approx 5{\times}10^5$ (Fig. 3); by the inverse function theorem the solution is locally unique. The singular spectrum spans roughly five orders of magnitude—from the speed directions (largest $\sigma$) down to the $d_W \leftrightarrow \delta$

geometry trade-off (smallest $\sigma$)—which is why first-order optimizers fail here ($\kappa(\boldsymbol{H}) \approx \kappa^2 \approx 10^{10}$) while trust-region Gauss–Newton succeeds, and why the certificate of Sec. V must not truncate weak directions: the weakest eigenvalue is genuine information, and it carries the dominant error bound. The classical speed condition $|N_i| \neq |N_j|$ remains necessary; its quantitative form is condition (i) below.

At the *spectral* stage, identifiability is information-theoretic. Let $\sigma(\hat{N}_i)$, $\sigma(\hat{c}_i)$ denote the Fisher standard errors of the fitted lattice model (Sec. V).

**Definition 1** (Spectral identifiability)**.** *A configuration is spectrally identifiable at observation time $T$ if*

- *(i)* pair resolvability*:* $\min_{i\neq j} \min(|N_i - N_j|, |N_i + N_j|) > 3\,\sigma_{ij}$*,*
- *(ii)* prism detectability*:* $|c_{\boldsymbol{e}_i}| > 5\,\sigma(\hat{c}_i)$ *for all $i$,*
- *(iii)* lattice non-degeneracy*:* $|\boldsymbol{k}\cdot\boldsymbol{N} - N_j| > 3\sigma$ *for all* $|\boldsymbol{k}|_1 \leq 4$*,* $\boldsymbol{k} \neq \pm\boldsymbol{e}_j$*,*
- *(iv) the TIR-masked sample fraction is below the mask budget,*
- *(v)* front-end capacity*: the number of lattice lines above the working floor does not exceed the line-model order of the spectral front end.*

Conditions (i)–(iv) are properties of the *signal*; (v) is a property of the *estimator*, and it is qualitatively different: its violation is self-diagnosed at run time (the CLEAN residual sits far above the lattice-adequacy floor of Theorem 1) and is curable *without additional observation* by switching to a capacity-free candidate source (Sec. IV B). Deeply non-paraxial systems reach this regime through a dense comb whose tooth spacing is the minimal small-$\boldsymbol{k}$ lattice value; past capacity a subspace line estimator returns cluster centroids rather than lines, poisoning every downstream decision while the information itself remains intact.

Conditions (i)–(iii) fail continuously, and each failure carries a prescription for the observation time that would cure it, $T_{\rm req}$ of Eq. (14), derived in Sec. V from the $T^{-3/2}$ frequency-error law and the estimator's resolution floor. The campaign's target is *zero unexplained failures*: every recovery failure must be a certified violation of (i)–(iv) (E1), and every prescription is validated by bisection (E4).

### D. Wedge count and non-Risley targets

Two classical facts settle the wedge-count question. The set $\bigcup_P \mathcal{Q}_P$ of quasi-periodic functions is dense in $L^2[0,T]$ (Stone–Weierstrass [28]), and for targets $f \in W^{s,2}$ the best-$P$-prism residual obeys $d(f,\mathcal{T}_P)^2 \leq C_s \|f\|^2_{W^{s,2}}\, P^{-(2s-1)}$ in the paraxial limit [29] (measured on a piecewise-linear target: $d^2 \sim P^{-1.64}$, MSE 20.2 at $P$=1 down to 0.93 at $P$=6). A true Risley pattern reaches $d = 0$ exactly at the true $P$: the prism count is identified as the $P$ whose residual is numerical zero, and a strictly positive residual plateau is itself a certificate that the target is not a Risley pattern of the assumed geometry.

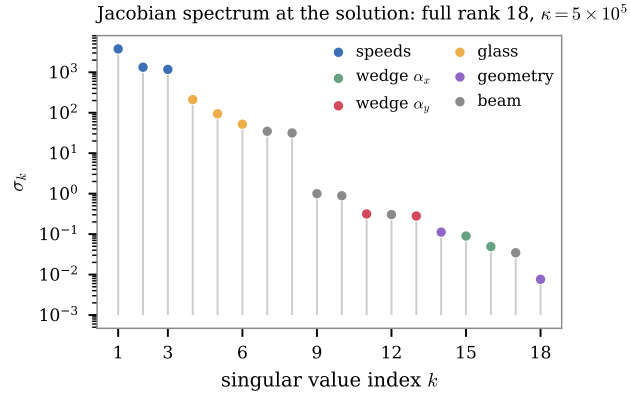


FIG. 3. Jacobian singular-value spectrum at the recovered solution of the Fig. 1 instance ($P$=3, 18 parameters; marker color indicates the dominant parameter group of each right singular direction). All 18 singular values are strictly positive (full rank: local identifiability), spanning five orders of magnitude from speeds to the $d_W \leftrightarrow \delta$ geometry trade-off. The weakest direction is genuine information and dominates the certified error bound (Sec. V).

## IV. THE INVERSION ALGORITHM

All stages operate on one sampled pattern ($T = 10\,$s, $f_s = 20\,$Hz, 200 samples in the reference acquisition). The spectral stages perform *no* forward-model evaluations; the angle stage performs six; the completion stage performs one bounded trust-region solve. Median wall time is $\sim$1 s (spectral) plus 1–4 s (completion) per instance.

### A. De-glitching the TIR set

Samples adjacent to pattern jumps exceeding 10× the median step are masked (with a 15% mask budget); the masked samples are excluded from every subsequent fit, and linearly interpolated only for the pencil initializer. After the final lattice fit the mask is recomputed self-consistently from the fit residual. The TIR cases of Prop. 2 are thereby handled, not excluded: the recovery rate conditional on a TIR crossing is reported separately in the atlas (**[E1: TIR class]**).

### B. CLEAN line growth at order one

The matrix pencil [20, 21] applied to the (masked, de-meaned) signal returns super-resolved line frequencies and complex amplitudes, free of the bin discretization that destroys FFT-based phase readout. Lines are accepted one at a time, CLEAN-style [24]: take the strongest pencil line of the current residual, add it to the line set, and refit *all* line frequencies jointly by variable projection [25] (amplitudes linear, frequencies by damped Gauss–Newton) at lattice order $B{=}1$, where a line explains only itself and Lemma 2 has no purchase. Novel lines—those not expressible as small-integer combinations of the current set—take priority, so harmonics cannot exhaust the line budget before a weak fundamental appears. If the CLEAN residual reports front-end overload [condition (v)], the growth is repeated once with capacity-free candidates—windowed-FFT peak maxima with parabolic interpolation—whose precision the joint Gauss–Newton refit then restores; the retry fires only on overloaded instances (development testing found it curing the entire overload class at the original observation time; the campaign measures the cure rate at scale, **[E1: overload class]** ).

### C. Generator identification by integer coverage

Candidate generator $P$-tuples among the fitted lines are ranked by *coverage*: the amplitude-weighted fraction of all lines representable as $\boldsymbol{k}\cdot\boldsymbol{g}$ with $|\boldsymbol{k}|_1 \le 3$ (sampling aliases $f \pm f_s$ included), minus a small penalty on representation order. A consensus reweighting zeroes lines that no leading candidate explains (CLEAN artifacts). Per Lemma 2, coverage—not fit residual—is the primary gate; the full lattice fit (order $B{=}3$, refined at $B{=}4$) referees only among candidates within a small coverage band, and a candidate whose generators drift far from the measured lines during refinement is rejected outright. Two deterministic repairs follow: a large-amplitude line at order $\ge 2$ is trial-promoted to a generator (it may be a fundamental sitting on an accidental near-relation), and if coverage reports unexplained lines, the strongest residual line is trial-swapped for each generator. Each repair is accepted only on a large, guarded residual improvement.

Two numerical protections in the lattice fit are load-bearing. Near-coincident lattice lines (closer than $0.012\,\mathrm{Hz}$ at $T = 10\,\mathrm{s}$) are *merged* onto their lowest-order representative—the data cannot distinguish them, and an unmerged pair lets ridge regression split large amplitudes across near-collinear columns; conversely the amplitude solve is *ridged*, because unregularized least squares on near-collinear columns manufactures huge canceling amplitude pairs that poison every amplitude-based decision. A final "sharp" pass with both protections off, from the converged basis, recovers the last decimal.

### D. Canonicalization and the sign test

The physical fundamentals are extracted from the fitted model as the rank-extending largest-amplitude lines (first order dominates: *Test A5* measures the violation rate at **[A5]** , all violations tolerated by the rank-extension rule); this single re-extraction can rescue a misselected generator. After the final refinement the generators *are* the speeds and are never re-extracted (from a refit, re-extraction grabs ridge-splitting twins—a documented failure of an earlier version). Finally the sign of each speed is verified by Prop. 3: if the fitted amplitude at $-\boldsymbol{e}_i$ exceeds that at $+\boldsymbol{e}_i$, the sign is flipped and the model refit. The test runs *unconditionally*—on every code path, from a dedicated fit if necessary: an implementation that proves the leak inequality but only sometimes enforces it will eventually return an estimate violating it by an order of magnitude (an ensemble case did exactly that before this was understood).

### E. Angle readout and completion

$\alpha_{y,i}$ is read from the fundamental phase (Prop. 4); the branch $|\arg c| > 90°$ simultaneously fixes $\operatorname{sign}\alpha_{x,i}$ (Prop. 1). $|\alpha_{x,i}|$ is obtained by inverting the cubic amplitude law (Prop. 5) with gains calibrated by two forward evaluations per prism—against the known geometry in the 9-parameter protocol, or against nominal mid-box geometry in the full 18-parameter problem. The beam angles are initialized analytically from the pattern's DC component (first-order rotating deflections average to zero, so the DC is the static ray); glass and distances start mid-box. A single bound-constrained trust-region solve [30] on the exact model, with TIR-masked residuals, completes the recovery to near machine precision.

If the pattern MSE does not reach the $10^{-12}$ acceptance threshold, a deterministic *verified ladder* is walked: gain recalibration using the first fit's geometry, the phase-branch flips of ambiguous prisms, a zero-angle start, a sign flip of the weakest speed (the least reliable bit of the extraction, cf. the A2 violation), and the alternate generator tuples ranked by the spectral stage. Every rung is a fixed candidate accepted or rejected by the final MSE—model selection, not search: the accepted solution reproduces the data to $10^{-23}$–$10^{-26}$, twelve orders below any rejected rung.

### F. Measured basin of the completion stage

The staged design rests on one empirical claim: spectral-quality initializations lie in the trust-region basin. We measure it directly (*Test A7*): perturbing true solutions by draws matched to the spectral stage's actual error scale (speeds $3\times10^{-4}\,\mathrm{Hz}$, $\alpha_x$ 1.5°, $\alpha_y$ 2°, glass 0.08, $d_W$ 40, gap 4, beam 2–3°), the 18-D trust-region solve converges in **[A7]** of trials; at 4× that error scale, still **[A7b]**. The historical "narrow basin" of this problem was a property of uninformed initializations, not of the landscape.

## V. ERROR ANALYSIS AND CERTIFICATES

This section derives the error bounds; nothing in it is asserted from simulation. The logical structure is: a completion-stage bound that is valid *whenever the residual test accepts, regardless of how the initializer arrived there* (Prop. 6); and a spectral-stage information analysis that governs *whether* the pipeline arrives, with closed-form scaling laws (Props. 7–9). The campaign of Sec. VII then *validates* the derived bounds rather than substituting for them.

### A. Completion stage: the certified bound

Let $F : \Theta \subset \mathbb{R}^{18} \to \mathbb{R}^m$ be the sampled (masked) forward map, $\boldsymbol{y} = F(\boldsymbol{\theta}^*) + \boldsymbol{\varepsilon}$ the observation, $\hat{\boldsymbol{\theta}}$ the accepted iterate, $\boldsymbol{J} = DF(\hat{\boldsymbol{\theta}})$ (full column rank by Sec. III C), and $\boldsymbol{r} = F(\hat{\boldsymbol{\theta}}) - \boldsymbol{y}$ the residual.

**Proposition 6** (Error decomposition). *If $F$ is $C^2$ on the segment $[\hat{\boldsymbol{\theta}}, \boldsymbol{\theta}^*]$ with $L_2 = \sup\|D^2F\|$ there, then*

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* = \underbrace{(\boldsymbol{J}^\top\boldsymbol{J})^{-1}\boldsymbol{J}^\top\boldsymbol{r}}_{\textit{optimality gap (computable)}} + \underbrace{(\boldsymbol{J}^\top\boldsymbol{J})^{-1}\boldsymbol{J}^\top\boldsymbol{\varepsilon}}_{\textit{noise term}} + \boldsymbol{\rho}, \tag{10}$$

*with $\|\boldsymbol{\rho}\| \le \frac{1}{2}L_2\|(\boldsymbol{J}^\top\boldsymbol{J})^{-1}\boldsymbol{J}^\top\|\,\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|^2$.*

*Proof.* Taylor with integral remainder gives $F(\boldsymbol{\theta}^*) = F(\hat{\boldsymbol{\theta}}) + \boldsymbol{J}(\boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}) + \boldsymbol{R}$, $\|\boldsymbol{R}\| \le \frac{1}{2}L_2\|\boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}\|^2$. Substituting into $\boldsymbol{r} = F(\hat{\boldsymbol{\theta}}) - F(\boldsymbol{\theta}^*) - \boldsymbol{\varepsilon}$ and applying $(\boldsymbol{J}^\top\boldsymbol{J})^{-1}\boldsymbol{J}^\top$ rearranges to (10) with $\boldsymbol{\rho} = (\boldsymbol{J}^\top\boldsymbol{J})^{-1}\boldsymbol{J}^\top\boldsymbol{R}$. □

The first term of (10) is exactly the remaining Gauss–Newton step: the certificate's gap term is an algebraic identity, not a heuristic—it vanishes at a stationary point and exactly accounts for early stopping. The second term is mean-zero with covariance $\sigma^2(\boldsymbol{J}^\top\boldsymbol{J})^{-1}$ under white noise of variance $\sigma^2$, and $s^2 = \|\boldsymbol{r}\|^2/(m-18)$ is its standard unbiased surrogate in the locally linear regime.

**Corollary 1** (Certificate). *Under $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2\boldsymbol{I})$ and local linearity ($\boldsymbol{\rho}$ subdominant),*

$$b_i \;=\; 3\big[s^2(\boldsymbol{J}^\top\boldsymbol{J})^{-1}\big]_{ii}^{1/2} \;+\; \big|(\boldsymbol{J}^\top\boldsymbol{J})^{-1}\boldsymbol{J}^\top\boldsymbol{r}\big|_i \tag{11}$$

*covers $|\hat{\theta}_i - \theta_i^*|$ with probability $\ge 0.997$ per parameter. In the noiseless numerical setting $\boldsymbol{\varepsilon}$ is the arithmetic-plus-truncation floor, the gap term dominates, and the bound is deterministic up to $\boldsymbol{\rho}$.*

Three remarks, each with teeth. (i) *No covariance truncation is admissible*: removing the smallest eigenpair of $\boldsymbol{J}^\top\boldsymbol{J}$ deletes precisely its largest contribution to $[(\boldsymbol{J}^\top\boldsymbol{J})^{-1}]_{ii}$, which lies along the $d_W \leftrightarrow \delta$ direction whose eigenvalue falls to $\sim 10^{-15}$ of the largest on the worst-conditioned ensemble instances—far below the typical $\sim 10^{-12}$ set by $\kappa \approx 5 \times 10^5$; rank 18 (Sec. III C) guarantees invertibility, and a conventional $10^{-14}$ pseudo-inverse cutoff measurably broke coverage on one development case before this was understood. (ii) *Path independence*: nothing in Prop. 6 references the initializer. Once the verified ladder accepts, the certificate holds even if every spectral heuristic had been wrong—the spectral stage controls only whether acceptance occurs. (iii) *Calibration is testable*: at $3\sigma$ the per-parameter miss rate should be $\approx 0.3\%$; the campaign's E2 experiment tests this against $\sim 10^7$ ground-truth checks (measured miss rate **[E2]**), resolving mis-calibration at the $10^{-4}$ level—the bounds must come out statistically calibrated, neither optimistic nor padded.

### B. Spectral stage: information limits

Model the analytic signal as $z_j = \sum_{\boldsymbol{k}\in K} c_{\boldsymbol{k}} e^{2\pi i(\boldsymbol{k}\cdot\boldsymbol{N})t_j} + w_j$ at $t_j = j/f_s$, $j = 0, \ldots, M-1$, $T = M/f_s$, with complex white noise of variance $\sigma_w^2$ (in noiseless data, the lattice-truncation floor of Theorem 1 plays this role).

**Lemma 3** (Uniform-grid kernel bound). *For $t_j = j/f_s$, $j = 0, \ldots, M-1$, an offset $0 < |\delta| \le f_s/2$, and any weight sequence $q(t_j)$ that is monotone in $j$ with range in $[0,1]$ (e.g. $q = (t/T)^m$, $m \le 2$),*

$$\Big|\sum_j q(t_j)\, e^{2\pi i\delta t_j}\Big| \;\le\; \frac{f_s}{|\delta|} \;=\; \frac{M}{|\delta|\,T}. \tag{12}$$

*Proof.* For $q \equiv 1$ the sum is geometric with modulus $|\sin(\pi\delta M/f_s)|/|\sin(\pi\delta/f_s)| \le 1/\sin(\pi|\delta|/f_s) \le f_s/(2|\delta|)$, using $\sin\pi x \ge 2x$ on $[0, \frac{1}{2}]$. For general monotone $q$ with range in $[0,1]$, Abel summation against the partial sums $S_J$ (each $|S_J| \le f_s/2|\delta|$) gives the bound $(\sup|q| + \mathrm{TV}(q))\max_J|S_J| \le 2 \cdot f_s/(2|\delta|)$. □

**Proposition 7** (Lattice Fisher information). *The Fisher matrix of $(\boldsymbol{N}, \mathrm{Re}\,\boldsymbol{c}, \mathrm{Im}\,\boldsymbol{c})$ under the complex-Gaussian model is* exactly $(2/\sigma_w^2)\,\mathrm{Re}(\boldsymbol{X}^H\boldsymbol{X})$ *with $\boldsymbol{X} = [\boldsymbol{J}_N, \boldsymbol{E}, i\boldsymbol{E}]$, where $\boldsymbol{E}$ is the Vandermonde design and $\boldsymbol{J}_N$ its derivative in the generators—the matrix the failure certificate inverts. Let $\Delta$ be the minimal separation of distinct occupied (folded) lattice frequencies, $w_{\max}$ the ratio of largest to smallest fitted line weight, and suppose $\varepsilon := C_0\, n_{\text{lines}}\, w_{\max}/(\Delta T) \le \frac{1}{2}$ for an absolute constant $C_0$. Then, with the generator Gram matrix*

$G_{il} = \sum_{\boldsymbol{k}} k_i k_l |c_{\boldsymbol{k}}|^2$,

$$\sigma(\hat{N}_i)^2 = \frac{3\,\sigma_w^2}{2\pi^2 f_s\, T^3}\,[G^{-1}]_{ii}\,\big(1+\mathcal{O}(\varepsilon)+\mathcal{O}(M^{-2})\big) \qquad (13)$$
$$\geq \frac{3\,\sigma_w^2}{2\pi^2 f_s\, T^3\, S_i}\,\big(1-\mathcal{O}(\varepsilon)\big),$$

*where $S_i = G_{ii} = \sum_{\boldsymbol{k}} k_i^2 |c_{\boldsymbol{k}}|^2$, with equality in the second line iff $G$ is diagonal (decoupled generators); the scaling laws below use the conservative diagonal evaluation, while the computed certificate always inverts the full matrix.*

*Proof. Exact form.* For $z_j = \mu_j + w_j$ with circular complex Gaussian noise the log-likelihood is $-\sigma_w^{-2}\sum_j |z_j - \mu_j|^2$ + const, whence $I_{ab} = (2/\sigma_w^2)\,\mathrm{Re}\sum_j \overline{\partial_a \mu_j}\,\partial_b \mu_j$; with $\partial \mu_j/\partial N_i = 2\pi i\, t_j \sum_{\boldsymbol{k}} k_i c_{\boldsymbol{k}} e^{2\pi i \omega_{\boldsymbol{k}} t_j}$ and amplitude columns $\boldsymbol{E}, i\boldsymbol{E}$ this is the stated matrix.

*Block structure.* Rescale the generators to $\nu_i = 2\pi N_i T$ so that every design column has entries bounded by its line weight times $(t_j/T)^m$, $m \leq 1$. Same-line inner products are exact moment sums $\sum_j (t_j/T)^{m+m'} = M/(m+m'+1) + \mathcal{O}(1)$; cross-line inner products carry $e^{2\pi i \delta t_j}$ with $|\delta| \geq \Delta$ and are bounded by $M/(\Delta T)$ times the two line weights (Lemma 3). Hence $I = (2M/\sigma_w^2)(\boldsymbol{D}+\boldsymbol{R})$ with $\boldsymbol{D}$ block-diagonal per line and $\|\boldsymbol{D}^{-1}\boldsymbol{R}\| \leq \varepsilon$; the Neumann series gives $I^{-1} = (\sigma_w^2/2M)\boldsymbol{D}^{-1}\big(1+\mathcal{O}(\varepsilon)\big)$.

*Schur complement.* Within the block of line $\boldsymbol{k}$, eliminating the two amplitude coordinates replaces the raw second moment $\sum_j (t_j/T)^2$ by the centered one, $\sum_j (t_j - \bar{t})^2/T^2 = \frac{M}{12}(1-M^{-2})$ exactly [$t_j$ is a uniform grid]; the factor multiplying it in the $(N_i, N_l)$ entry is $(2\pi T)^2 k_i k_l |c_{\boldsymbol{k}}|^2$. Summing over lines, the generator block of $I$ after elimination is $(2/\sigma_w^2)(2\pi T)^2 \frac{M}{12}\, G\,(1+\mathcal{O}(\varepsilon) + \mathcal{O}(M^{-2}))$; inverting and using $M = f_s T$ gives (13). The final inequality is $[G^{-1}]_{ii} \geq 1/G_{ii}$, valid for every positive-definite $G$: $1 = (\boldsymbol{e}_i^\top G^{1/2} G^{-1/2}\boldsymbol{e}_i)^2 \leq (\boldsymbol{e}_i^\top G \boldsymbol{e}_i)(\boldsymbol{e}_i^\top G^{-1}\boldsymbol{e}_i)$ by Cauchy–Schwarz. The single-line special case recovers the classical single-tone bound [27]. ☐

**Corollary 2** (Observation-time law)**.** *At fixed sampling rate, $\sigma(\hat{N}_i) \propto T^{-3/2}$; a separation margin deficient by the factor $3\sigma/\Delta$ is closed at $T_{\mathrm{stat}} = T\,(3\sigma/\Delta)^{2/3}$.*

**Proposition 8** (Pair degradation and the resolution floor)**.** *(i) For two unit-weight lines at gap $g$ with $g/f_s \leq \frac{1}{2}$, the normalized column inner product has modulus $|\rho(g)| = \sin(\pi g T)\,/\,\big(M \sin(\pi g/f_s)\big)$ exactly, and for $gT \leq 1$,*

$$|\rho|^2 = 1 - \tfrac{\pi^2}{3}(gT)^2\big(1-M^{-2}\big) + \mathcal{O}\big((gT)^4\big);$$

*consequently the $2\times 2$ amplitude Gram matrix $M\left(\begin{smallmatrix} 1 & \rho \\ \bar{\rho} & 1\end{smallmatrix}\right)$ has smallest eigenvalue $M(1-|\rho|) = \Theta(M(gT)^2)$, and the least-squares variance of the amplitude-difference coordinate inflates by the factor $(1-|\rho|^2)^{-1} = \Theta\big((gT)^{-2}\big)$ relative to the separated case. (ii) Independently, the estimator merges lines closer than $C_{\mathrm{m}}/T$ ($C_{\mathrm{m}} = 0.12$; Sec. IV C)—a deliberate trade of super-resolution for robustness of the amplitude solve—so a pair or lattice-relation gap $g$ is resolvable only if $T \geq c\,C_{\mathrm{m}}/g$, and the certified prescription is*

$$T_{\mathrm{req}} \;=\; \max\Big(T\,(3\sigma/g)^{2/3},\; c\,C_{\mathrm{m}}/g\Big), \qquad c \approx 2. \quad (14)$$

*Proof.* (i) The inner product of the columns $(e^{2\pi i f t_j})_j$ and $(e^{2\pi i (f+g) t_j})_j$ is the geometric sum $\sum_j e^{2\pi i g t_j} = e^{i\pi g(M-1)/f_s}\,\sin(\pi g M/f_s)/\sin(\pi g/f_s)$; dividing by $M$ and using $T = M/f_s$ gives $|\rho|$. Apply $\sin x = x(1 - x^2/6 + \mathcal{O}(x^4))$ to numerator ($x = \pi g T$) and denominator ($x = \pi g/f_s = \pi g T/M$) and square. The eigenvalues of the Hermitian Gram form are $M(1 \pm |\rho|)$, and $1 - |\rho| = \frac{1}{2}(1-|\rho|^2)(1+\mathcal{O}((gT)^2))$. Explicit $2\times 2$ inversion of the normal equations shows the variance of the difference coordinate carries the factor $(1-|\rho|^2)^{-1}$. (ii) is the definition of the merge step; the safety constant $c$ is measured, not derived (Sec. V C). ☐

The second argument of (14) was forced on us by development-scale testing: in an ensemble of several hundred configurations, every unsolved lattice-relation case had $g = 1.4$–$3.2\,\mathrm{mHz}$, satisfying the statistical term but violating the merge floor ($cC_{\mathrm{m}}/g = 75$–$170\,\mathrm{s}$, beyond the $80\,\mathrm{s}$ ladder of that experiment)—the corrected formula classifies all of them. E4 re-tests the corrected prescription at campaign scale.

**Proposition 9** (Detectability)**.** *If all pairwise separations of occupied lines are $\geq \Delta$ and $n_{\mathrm{lines}}/(\Delta T) \leq \frac{1}{2}$, then $\mathrm{var}(\hat{c}_{\boldsymbol{k}}) = (\sigma_w^2/M)\,\big(1 + \mathcal{O}(n_{\mathrm{lines}}/(\Delta T))\big)$, so the $5\sigma$ detection test requires $|c_{\boldsymbol{e}_i}| \gtrsim 5\sigma_w/\sqrt{f_s T} \propto T^{-1/2}$; through the amplitude law (Prop. 5) the minimum detectable wedge magnitude is $\alpha_{\min} = \arctan\big(5\,\sigma(\hat{c})/a_i\big)$, which is the figure the weak-prism certificate reports.*

*Proof.* The amplitude Gram matrix is $M(\boldsymbol{I}+\boldsymbol{R})$ with $|R_{\boldsymbol{k}\boldsymbol{k}'}| \leq 1/(\Delta T)$ for $\boldsymbol{k} \neq \boldsymbol{k}'$ (Lemma 3), so $\|\boldsymbol{R}\| \leq n_{\mathrm{lines}}/(\Delta T) \leq \frac{1}{2}$ and the least-squares covariance is $\sigma_w^2[(\boldsymbol{X}^H\boldsymbol{X})^{-1}]_{\boldsymbol{k}\boldsymbol{k}} = (\sigma_w^2/M)(1+\mathcal{O}(\|\boldsymbol{R}\|))$ by the Neumann series; then $M = f_s T$, the $5\sigma$ rule, and inversion of the (odd, monotone-near-zero) amplitude law give the threshold. ☐

### C. What is derived, what is measured

The bound (11) and the laws (13)–(14) are analysis; two links in the chain are irreducibly empirical and are measured rather than assumed: the basin of the completion stage (Test A7—no convexity theory is available for this landscape, so we measure the basin at the actual init-error scale), and the constant $c$ in the merge floor. Every other constant in the certificates is computed from the data instance at hand.

## VI. SCALING THEORY FOR ARBITRARY PRISM COUNT

Everything so far holds for any $P$: Theorem 1 and Props. 3–5 are per-prism statements, the algorithm of Sec. IV takes the generator count as a parameter, and the certificates of Sec. V are dimension-agnostic. What changes with $P$ is the *information geometry*: the unknown vector grows linearly, $\dim\boldsymbol{\theta} = 4P+6$ (four per-prism parameters—speed, two wedge components, index—plus six global ones), while the spectral population it must be read from grows polynomially and its margins shrink. This section makes those statements theorems. Throughout, the speed magnitudes $m_i = |N_i|$ are i.i.d. uniform on $[\nu, W]$ (the admissible box), $L = W - \nu$, signs independent, and the working lattice orders are fixed integers: $B$ for fitting and $K$ for relation screening (the implementation uses $B$=3, $K$=4).

### A. Cost is polynomial in $P$

**Lemma 4** (Lattice population). *Let $\mathcal{K}(P,B) = \{\boldsymbol{k} \in \mathbb{Z}^P : 0 < |\boldsymbol{k}|_1 \le B\}$. Then, for fixed $B$,*

$$|\mathcal{K}(P,B)| \;=\; \sum_{j=1}^{\min(P,B)} 2^j \binom{P}{j}\binom{B}{j} \;=\; \Theta\big(P^B\big). \tag{15}$$

*Proof.* A vector with exactly $j$ nonzero entries is determined by its support ($\binom{P}{j}$ choices), the signs of those entries ($2^j$), and the magnitudes, i.e. an integer composition $m_1 + \cdots + m_j \le B$ with $m_i \ge 1$, of which there are $\binom{B}{j}$. Summing over $j$ gives the identity; for $P > B$ the $j = B$ term dominates, giving $2^B\binom{P}{B} = \Theta(P^B)$. □

Hence the lattice design matrices have $\Theta(P^B)$ columns, the completion Jacobian is $2M\times(4P+6)$, and every stage costs polynomial time in $P$ (tuple selection ranks candidates by amplitude and coverage; it never enumerates exhaustively). The *readout* is strictly linear in $P$: orientations are $P$ line phases, magnitudes invert $P$ scalar cubics calibrated by $2P$ forward evaluations. Nothing in the algorithm is exponential in $P$; the obstructions below are informational, not computational.

### B. Crowding: pair resolvability costs $\Theta(P^2)$

**Proposition 10** (Exact crowding law). *Let $g_{\mathrm{pair}} = \min_{i\ne j}\min(|N_i - N_j|, |N_i + N_j|)$. For $s < 2\nu$,*

$$\Pr\big[g_{\mathrm{pair}} > s\big] = \Big(1 - \frac{(P-1)\,s}{L}\Big)_+^P, \quad \mathbb{E}\,g_{\mathrm{pair}} = \frac{L}{P^2-1}. \tag{16}$$

*Proof.* Whichever branch of $\min(|N_i - N_j|, |N_i + N_j|)$ is not $|m_i - m_j|$ equals $m_i + m_j \ge 2\nu$, so for $s < 2\nu$ the pair gap is the minimum spacing of the $P$ i.i.d. magnitudes on $[\nu, W]$. Conditioning on the order statistics, the $P$+1 spacings (two edge gaps, $P$−1 interior gaps) are uniform on the simplex $\{d_j \ge 0,\ \sum d_j = L\}$. Subtracting $s$ from each interior gap maps the event {all interior gaps $> s$} affinely onto the same simplex with total length $L - (P-1)s$; uniform measure on a $P$-dimensional simplex scales as the $P$-th power of its linear size, giving the survival function. Integrating it, $\int_0^{L/(P-1)}\big(1 - \frac{(P-1)s}{L}\big)^P ds = \frac{L}{(P-1)(P+1)}$. □

**Corollary 3** (Crowding-limited observation time). *By the merge floor (Prop. 8), resolving all pairs requires $T \ge c\,C_{\mathrm{m}}/g_{\mathrm{pair}}$; by* (16), $\Pr[g_{\mathrm{pair}} < s] \le P(P-1)s/L$, *so condition (i) holds with probability $\ge 1-\delta$ once*

$$T \;\ge\; T_{\mathrm{pair}}(P,\delta) \;=\; \frac{c\,C_{\mathrm{m}}\,P(P-1)}{L\,\delta} \;=\; \Theta\Big(\frac{P^2}{\delta}\Big). \tag{17}$$

### C. Accidental relations: a degree-$K$ demand

Condition (iii) requires the fundamentals to avoid the rest of the lattice. Writing coincidences $\boldsymbol{k}\cdot\boldsymbol{N} \approx N_j$ as relations $(\boldsymbol{k} - \boldsymbol{e}_j)\cdot\boldsymbol{N} \approx 0$, the governing quantity is $g_{\mathrm{rel}} = \min\{|\boldsymbol{k}\cdot\boldsymbol{N}| : \boldsymbol{k} \in \mathcal{K}(P,K),\ \boldsymbol{k} \ne \pm q\boldsymbol{e}_j\}$.

**Proposition 11** (Relation-gap anti-concentration). *For every $\varepsilon < \nu$,*

$$\Pr\big[g_{\mathrm{rel}} < \varepsilon\big] \;\le\; \frac{2\,\varepsilon}{L}\,\big|\mathcal{K}(P,K)\big| \;=\; \Theta\Big(\frac{\varepsilon\,P^K}{L}\Big). \tag{18}$$

*Moreover the expected number of relations below $\varepsilon$ is also $\Theta(\varepsilon P^K/L)$, which is $\Theta(1)$ at $\varepsilon \sim LP^{-K}$: no better polynomial rate is available from first moments, and the campaign measures the empirical law of $g_{\mathrm{rel}}$ directly ( **[E1: margin histograms]** ).*

*Proof.* Fix $\boldsymbol{k}$ with at least two nonzero entries and let $i^*$ index a largest-magnitude entry, $1 \le |k_{i^*}| \le K$. Conditioning on all coordinates except $m_{i^*}$ (signs are fixed within a configuration), the map $m_{i^*} \mapsto \boldsymbol{k}\cdot\boldsymbol{N}$ is affine with slope of modulus $|k_{i^*}| \ge 1$, so the conditional density of $\boldsymbol{k}\cdot\boldsymbol{N}$ is bounded by $1/(|k_{i^*}|L) \le 1/L$ and $\Pr[|\boldsymbol{k}\cdot\boldsymbol{N}| < \varepsilon] \le 2\varepsilon/L$. Single-support vectors $\pm q\boldsymbol{e}_j$ are excluded, and for them $|q\,m_j| \ge \nu > \varepsilon$ anyway. The union bound over $\mathcal{K}(P,K)$ with Lemma 4 gives (18); linearity of expectation over the same family, with each term also bounded *below* by $c'\varepsilon/L$ for the $\Theta(P^K)$ vectors whose value range straddles zero, gives the expected count. □

**Corollary 4** (Relation-limited observation time). *Setting $\varepsilon = c\,C_{\mathrm{m}}/T$ in* (18), *condition (iii) holds with probability $\ge 1-\delta$ once*

$$T \;\ge\; T_{\mathrm{rel}}(P,\delta) \;=\; \Theta\Big(\frac{C_{\mathrm{m}}\,P^K}{L\,\delta}\Big), \tag{19}$$

*polynomial of degree $K$—the dominant demand once $K > 2$.*

**Proposition 12** (Pigeonhole at unbounded order)**.** *For any $\boldsymbol{N} \in [-W, W]^P$ and any integer $Q \geq 1$ there exists $\boldsymbol{k} \neq \boldsymbol{0}$, $|\boldsymbol{k}|_\infty \leq Q$, with*

$$|\boldsymbol{k} \cdot \boldsymbol{N}| \;\leq\; \frac{P\,Q\,W}{(Q+1)^P - 1}. \tag{20}$$

*Proof.* The $(Q+1)^P$ values $\{\boldsymbol{k} \cdot \boldsymbol{N} : \boldsymbol{k} \in \{0, \ldots, Q\}^P\}$ lie in an interval of length $PQW$; two are within $PQW/((Q+1)^P - 1)$ of each other, and their difference is the required vector. □

Equation (20) is exponentially small in $P$: if the working order ever had to *grow* with $P$, near-relations would become unavoidable and identifiability would collapse. The saving fact is that it does not: the geometric coefficient decay of Theorem 1 confines the observable spectrum to bounded order, so the demand stays polynomial (19). This is the precise sense in which the large-$P$ problem is Diophantine, and it identifies the bounded-order window as the load-bearing structural assumption.

### D. Capacity and detectability

**Proposition 13** (Capacity threshold)**.** *For every $\boldsymbol{N}$, some pair of the folded lattice frequencies $\{\boldsymbol{k} \cdot \boldsymbol{N} \bmod f_s : \boldsymbol{k} \in \mathcal{K}(P, B)\}$ lies within $f_s/|\mathcal{K}(P, B)|$ of each other. Hence once $T \gtrsim (C_{\mathrm{m}}/f_s)\,|\mathcal{K}(P, B)| = (C_{\mathrm{m}}/f_s)\,\Theta(P^B)$, some lattice spacing lies below the merge floor for* every *configuration: the dense-comb regime of condition (v) is generic rather than exceptional, and the front end must run in its capacity-free mode (Sec. IV B).*

*Proof.* Pigeonhole: $|\mathcal{K}(P, B)|$ points on a circle of circumference $f_s$ contain a pair at arc distance $\leq f_s/|\mathcal{K}(P, B)|$; compare with $C_{\mathrm{m}}/T$ and apply Lemma 4. □

The campaign locates the threshold empirically by the overload-retry rate as a function of $P$ ( **[E5]** ).

**Proposition 14** (Detectability decays at most geometrically in $P$)**.** *Let $D_{\min} \leq D_m \leq D_{\max}$ bound the per-interface transfer derivatives of Lemma 1 over the admissible box, with $D_{\min} \geq (n_{\min}/n_{\max}) \cos\theta_{\max} > 0$. Then the paraxial gain of prism $i$ obeys*

$$\begin{aligned} K_{\bullet,i} \;&\geq\; d_W\, D_{\min}^{2(P-i)}, \\ K_{\bullet,i} \;&\leq\; \big(d_W + (P{-}i)(\tau_{\mathrm{p}}{+}\delta)\big) \sec^2\!\theta_{\max}\, D_{\max}^{2(P-i)}, \end{aligned} \tag{21}$$

*where $\tau_{\mathrm{p}}$ is the prism thickness. Hence $\log a_i$ [Prop. 5] varies at most linearly in $P$ with slope bounded by $2 \log D_{\max}$; at zero beam angles every $\theta = 0$, each prism traversal contributes $D = (1/n_{\mathrm{g}}) \cdot n_{\mathrm{g}} = 1$ exactly, and the gains are $P$-independent up to the linear growth of the optical path. Condition (ii) therefore degrades at most geometrically in $P$ with an explicitly bounded rate, while the lattice population grows polynomially (Lemma 4): for moderate $P$ the binding constraints are crowding and relations, not detectability.*

*Proof.* For the lower bound keep only the final summand of (7)—the workpiece segment of length $d_W$, reached after the $2(P-i)$ downstream interfaces—and bound each transfer derivative below by $D_{\min}$. For the upper bound, bound every segment length by the total downstream path, every secant by $\sec^2\theta_{\max}$, and every derivative by $D_{\max}$. The value of $D$ at zero angles follows from $D_m = n_{m^-} \cos\theta_{m^-}/(n_{m^+} \cos\theta_{m^+}) = n_{m^-}/n_{m^+}$, whose product over the entry–exit pair of one prism is $(1/n_{\mathrm{g}}) \cdot n_{\mathrm{g}} = 1$. □

The campaign measures the actual decay of median $|c_{\boldsymbol{e}_i}|$ with $P$ ( **[E5]** ).

### E. The feasibility frontier

**Theorem 2** (Feasibility scaling)**.** *Let $T^*(P, \delta)$ be the least observation time at which a random configuration satisfies conditions (i) and (iii) of Definition 1 with probability at least $1 - \delta$. For a fixed box and fixed working orders $(B, K)$,*

$$T^*(P, \delta) = \mathcal{O}\Big(\tfrac{1}{\delta} P^{\max(2,K)}\Big) \quad \textit{and} \quad T^*(P, \delta) = \Omega\big(P^2\big). \tag{22}$$

*Proof.* The upper bound is Corollaries 3 and 4 with the union bound over the two conditions. For the lower bound, take $s = \lambda L/P^2$ in (16): the survival probability tends to $e^{-\lambda}$, so with probability bounded away from zero $g_{\mathrm{pair}} \leq \lambda L/P^2$, and the merge floor (Prop. 8) converts this to $T \geq c\, C_{\mathrm{m}} P^2/(\lambda L)$. □

**Conjecture 1** (Empirical frontier exponent)**.** *On the campaign box the measured frontier follows $T^* \propto P^\gamma$ with $2 \leq \gamma \leq K$; we conjecture the crowding exponent $\gamma = 2$ governs the practical range $P \leq 6$, with relations taking over only beyond it. E5 fits $\gamma =$* ***[E5 fit]*** .

Two statements are deliberately left as measurements rather than theorems. First, recovery probability should be a function of the *margin coordinates* (pair gap, relation gap, minimal wedge) alone, with $P$ acting only through their distributions—a collapse prediction the atlas tests directly ( **[E1/E5 collapse]** ). Second, full column rank of the $(4P+6)$-column completion Jacobian at generic recovered points is verified per instance by the campaign ( **[E1: rank statistics]** ); we conjecture, but do not prove, that it holds for almost every configuration at every $P$.

## VII. THE LARGE-SCALE COMPUTATIONAL CAMPAIGN

The statistical evidence for this paper is a computational campaign sized to the claims: order $10^6$ random configurations under an adaptive observation protocol, order $10^7$ ground-truth certificate checks, and a prism-count frontier to $P$=6, executed as embarrassingly parallel single-core tasks (estimated 50–100 core-days). **Every yellow-boxed value and every framed figure in this section is a placeholder**, specified in advance of the runs; the analysis scripts that will fill them (`experiments/sweep.py`, `experiments/aggregate.py`) are frozen in the repository together with the exact statistical procedures described below, so the campaign is pre-registered in the strict sense: the plots and tables were designed before the data existed.

### A. Design and provenance

*Case stream.* Configuration $i \in \{0, 1, 2, \dots\}$ is generated by a counter-based pseudorandom generator keyed by $(\mathrm{seed}_0, i)$, uniform over the admissible box with $|N_i| \geq \nu$. Generation is *order-free*: any shard can produce any index, so the ensemble is defined by the integer range alone, and adding shards never changes existing cases. Results are append-only records, one per (configuration, condition), resume-safe by key.

*Margins at truth.* For every configuration the identifiability coordinates of Definition 1 are computed from ground truth and stored with the record: minimal pair separation $g_{\mathrm{pair}}$, minimal relation gap $g_{\mathrm{rel}}$ at order $K$, minimal wedge magnitude $\min_i |\alpha_{x,i}|$, and the cycle count of the slowest prism. Every rate reported below can therefore be resolved against the margins the theory says govern it—the difference between measuring a phase diagram and quoting an average.

*Adaptive protocol.* The unit of evaluation is the protocol, not a fixed $T$: solve at $T$=10 s; on failure, follow the certificate's prescription up the ladder $T$=20, 40, 80 s. Success is $\max_i |\hat{\theta}_i - \theta_i^*| < 10^{-3}$ (achieved errors are typically $10^{-11}$; the threshold only classifies).

*Statistics.* All rates carry 95% Wilson intervals; binned margin curves require ≥20 configurations per bin; exponent fits are ordinary least squares on log–log medians with bootstrap confidence intervals; nothing is truncated silently (any bounded sub-analysis states its bound).

*Arithmetic.* Single-threaded pinned BLAS throughout: multithreaded eigensolvers are not bitwise reproducible, and the CLEAN greedy amplifies the difference on boundary cases.

### B. E1: the identifiability atlas

$10^6$ configurations at $P$=3 under the adaptive protocol, with success certificates on every recovery and failure certificates on every miss. Reported: recovery probability at each ladder rung; the certificate-class decomposition of every failure, each class checked against ground truth (does the named deficit actually hold at truth?); the fraction of failures with *no* certificate—the zero-unexplained target, the campaign's sharpest falsifiable claim; and recovery probability as a function of the true margins with the derived thresholds overlaid.

**PENDING CAMPAIGN DATA**

**Figure E1 — the atlas (empty axes, to be filled)**

three log-$x$ panels, $y$ = recovery probability at $T$=10 s (%) with Wilson error bars per bin (≥20 cases/bin). (a) $x = g_{\mathrm{pair}}$ (Hz); overlay: vertical merge-floor line $C_{\mathrm{m}}/T = 0.012$ Hz. (b) $x = g_{\mathrm{rel}}$ (Hz); same overlay. (c) $x = \min_i |\alpha_{x,i}|$ (deg); overlay: derived detectability threshold $\alpha_{\min}(T)$ of Prop. 9. data: `sweep_adaptive_atlas_*.jsonl`; generator: `aggregate.py --figs`.

FIG. 4. The identifiability atlas: measured recovery probability against the three margin coordinates, with the *derived* phase boundaries overlaid. The prediction under test: recovery collapses to sharp transitions at the theoretical thresholds, and the transition locations are parameter-free (no fits).

**PENDING CAMPAIGN DATA**

**Table E1 — adaptive recovery, $10^6$ configurations**

rows: recovered at $T$=10 s; cured at 20/40/80 s; recovered, adaptive (total); certified infeasible at $T \leq 80$ s by class (close pair / relation / weak prism / TIR / rank-deficient); unexplained (target 0). columns: count, share, 95% Wilson interval. plus: median parameter error, median wall time, overload-retry rate and cure rate, TIR-crossing rate and conditional recovery.

### C. E2: certificate calibration at scale

Every recovered configuration is certified [Eq. (11)] and every bound is checked against ground truth. With $\sim 18$ parameter checks per configuration this is order $10^7$ Bernoulli trials of the $3\sigma$ claim (nominal miss rate 0.3% per parameter): the binomial test resolves mis-calibration at the $10^{-4}$ level, far beyond what any development-scale ensemble could see. Tightness (bound/true-error ratio) is reported as quantiles; the failure-certificate side is scored by class-conditional truth verification.

**PENDING CAMPAIGN DATA**

**Table E2 — calibration**

coverage count / total checks and Wilson interval vs. nominal 99.7%; per-parameter-group miss rates (speeds / angles / glass / geometry / beam); median and p90 tightness; failure-certificate truth-verification rate by class.

### D. E3: scaling-law exponents

Fixed-$T$ cohorts at $T \in \{5, 10, 20, 40, 80, 160, 320\}$ s (no early stopping, so every cohort is a complete sam-

**PENDING CAMPAIGN DATA**

**Figure E2 — bound vs. error (empty axes, to be filled)**

log–log density scatter, $x$ = true per-case maximal error, $y$ = certified maximal bound, $\sim 10^6$ points as a 2-D histogram; overlay: identity line (coverage = mass above it) and median-tightness guide. data: E1 records with `--bounds`; generator: `aggregate.py --figs`.

FIG. 5. Certificate calibration at scale: every certified bound against its ground-truth error. Predictions under test: mass above the identity at the nominal $3\sigma$ rate; tightness stable across orders of magnitude of error.

ple). The derived laws enter as *point predictions* on log–log fits, not free parameters: the frequency-error law $\sigma(\hat{N}) \propto T^{-3/2}$ [Eq. (13)], pair-gap variance inflation $\Theta((gT)^{-2})$ (Prop. 8), and amplitude detectability $\propto T^{-1/2}$ (Prop. 9).

**PENDING CAMPAIGN DATA**

**Figure/Table E3 — fitted exponents (empty axes, to be filled)**

per law: log–log plot of the median measured quantity vs. $T$ with OLS slope and bootstrap CI, predicted slope drawn as a guide line. table rows: law, derived exponent ($-3/2$, $-2$, $-1/2$), fitted exponent $\pm$ CI, verdict. data: `sweep_fixedT_*.jsonl`.

### E. E4: prescription validation

For every configuration unsolved at $T$=10 s, the minimal recovering observation time $T_{\min}$ is located by geometric bisection (capped at 320 s) and compared with the certificate's prescription $T_{\text{req}}$ [Eq. (14)]. The claim under test is one-sided validity with bounded conservatism: recovery at $T \le T_{\text{req}}$, with the sharpness measured by quantiles of $T_{\text{req}}/T_{\min}$.

**PENDING CAMPAIGN DATA**

**Figure/Table E4 — prescriptions (empty axes, to be filled)**

scatter $x = T_{\text{req}}$ (prescribed), $y = T_{\min}$ (bisected), log–log with identity line; validity = mass below the identity. table: median/p90/max $T_{\min}$; fraction with $T_{\min} \le T_{\text{req}}$; quantiles of $T_{\text{req}}/T_{\min}$; infeasible-below-cap count. data: `sweep_mint_*.jsonl`.

### F. E5: the prism-count frontier

Signed-speed extraction for $P$ = 2–6 on a $P \times T$ grid, $T \in \{10, 20, 40, 80\}$ s, margins recorded as in E1. This is the direct test of Sec. VI: (a) the margin-collapse prediction—recovery probability is a function of the margin coordinates, independent of $P$ given the margins; (b) the frontier exponent of Conjecture 1; (c) the capacity threshold of Prop. 13, located by the overload-retry rate; (d) the at-most-geometric gain-decay bounds of Prop. 14.

**PENDING CAMPAIGN DATA**

**Figure/Table E5 — the $P \times T$ frontier (empty axes, to be filled)**

(a) heat map: recovery rate over the $P \times T$ grid with per-cell intervals. (b) log–log $T^*(P)$ at fixed recovery level (e.g. 90%) with guide slopes $P^2$ and $P^K$; fitted $\gamma$. (c) margin-collapse check: recovery vs. $g_{\text{pair}}T$ pooled across $P$, one curve per $P$, prediction: curves coincide. (d) overload-retry rate vs. $P$; median $|c_{e_i}|$ vs. $P$. data: `sweep_speedsP_*.jsonl`.

### G. E6: noise

The adaptive protocol under additive white measurement noise at SNR $\in \{60, 50, 40, 30, 20\}$ dB. The certificate framework absorbs noise through the residual [$s^2$ in Eq. (11)]; the claims under test are that coverage survives the noise-driven inflation of the bounds (E2's calibration, repeated per SNR) and that precision is lost along the weak $d_W \leftrightarrow \delta$ singular direction first, exactly as the spectrum of Fig. 3 predicts.

**PENDING CAMPAIGN DATA**

**Figure/Table E6 — noise phase diagram (empty axes, to be filled)**

(a) heat map: recovery rate over SNR $\times$ $T$. (b) per-SNR coverage vs. nominal (as E2). (c) median bound by parameter group vs. SNR, prediction: geometry inflates first and fastest along $d_W \leftrightarrow \delta$. data: `sweep_noise_*.jsonl`.

### H. E7: baselines under identical conditions

The method is scored against three baselines on a common $10^4$-configuration subset of the atlas: the FFT peak-picking front end (magnitudes only; sign-blind by construction) that anchored all previous approaches, and our two strongest predecessor solvers. Since one of those is machine-learned we specify its design completely: it is a serious, physics-informed architecture, and its failure mode is what motivated the present method.

*Staged neural solver.* The learned solver factors the map along the same physical hierarchy as the lattice method. Stage 1 takes speed magnitudes from the top-$P$ FFT peaks and resolves the $2^P$ signs by screening every combination through the forward model seeded with stage-2 angles. Stage 2 is a fully connected network $28 \to 256 \to 256 \to 128 \to 6$ (batch normalization, GELU) mapping the three signed speeds plus a 25-dimensional spectral feature vector—top-3 peak frequencies, per-axis peak amplitudes, sin / cos of the peak phases, pattern centroid and spread—to the six box-normalized wedge angles through a sigmoid head. Stage 3 predicts the remaining nine parameters (glass, geometry, beam) by fusing a four-stage strided 1-D convolutional encoder over the raw $2 \times 200$ trace (channels $2 \to 32 \to 64 \to 128 \to 128$, kernels 7/5/3/3, global average pooling) with a conditioning branch on the 34 previously determined quantities. Both networks were trained on $10^6$ simulated patterns ($5 \times 10^4$ held out) drawn uniformly from the admissible box, with teacher-forced conditioning: AdamW at learning rate $10^{-3}$, weight decay $10^{-4}$, one-cycle schedule, batch 512, gradient clipping, 80 and 100 epochs respectively. The stage-3 loss augments parameter mean-square error with a physics-informed term—pattern MSE through a differentiable port of the forward model (validated to $2 \times 10^{-6}$ against the exact model), weight 0.1, evaluated every fourth batch. At inference the prediction receives 300 Adam refinement steps through the differentiable model before the same trust-region completion every method in the comparison shares.

Its failure rate ( **[E7]** ) is structural, not capacity- or data-limited. First, stage 1 inherits every pathology of the FFT front end: harmonics and cross-terms displace fundamentals among the top peaks in exactly the hard cases, corrupting the conditioning features downstream. Second, the completion basin is 1–5° wide in wedge angle (Test A7) while the angle posterior conditioned on corrupted peaks is multimodal; a regression network returns a compromise between modes that lies in no basin. More data or parameters removes neither obstruction; replacing learned initialization with the exact lattice coordinates removes both.

*Grid solver.* The strongest pre-lattice method replaces the networks with enumeration: candidate speed triples from the top-12 FFT peaks, ranked per triple by a coarse $5^3$ forward screen; for each of the best five triples, a $9^3$ joint grid over the three $\alpha_x$ (with $\alpha_y = 0$; the basin is wide in $\alpha_y$, narrow in $\alpha_x$), screened through the differentiable model, best nodes finished by trust region. Its failures are the FFT front end failing, not the grid: an oracle-speed control (identical grid, true speeds supplied) isolates the front end's share of the failure rate ( **[E7: oracle control]** ).

**PENDING CAMPAIGN DATA**

**Table E7 — baselines, common $10^4$-configuration subset**

rows: FFT top-$P$ front end (signed-speed success; sign-blind by construction); staged neural + trust region (9-D and 18-D); $\alpha_x$-grid + trust region (9-D); grid with oracle speeds (control); lattice inversion (9-D and full 18-D). columns: success rate with Wilson interval, median time per case, forward-model evaluations per case.

TABLE I. Assumption verification suite at campaign scale (all "measured" entries pending).

| claim | | measured | status |
|---|---|---|---|
| A1 | flip symmetry exact | **[max $\lvert\Delta F\rvert$]** | [–] |
| A2 | leak < main line | **[med. ratio; viol.]** | [–] |
| A3 | phase readout law | **[med.; p90]** | [–] |
| A4 | lattice support | **[med. residual]** | [–] |
| A5 | fundamental dominance | **[violation rate]** | [–] |
| A6 | amplitude law | **[med. (nominal)]** | [–] |
| A7 | completion basin | **[rate at scale]** | [–] |
| A8 | $T$-prescriptions | **[validity (E4)]** | [–] |

### I. E8: the assumption audit at scale

Every structural claim used anywhere in the method is verified by a named experiment over the campaign ensemble (Table I): A1–A4 and A6 are marginal measurements over the atlas records; A5 and A7 need dedicated instrumentation (fundamental-dominance ranking inside the fitted models; basin trials perturbing truth at the spectral stage's actual error scale); A8 is scored by E4. The audit's role is adversarial: a row that fails at scale falsifies the corresponding proposition's empirical premise, and the paper's claims must be revised—not the row deleted.

## VIII. DISCUSSION

### A. Relation to prior methods

The spectral stage assembles classical components—Prony-family line estimation [20, 21], CLEAN deconvolution [24], variable projection [25], and the parsimony principle of integer-relation finding [26]—around one problem-specific observation: for Risley systems the unknown frequencies are not free but lattice-generated, and the physically meaningful unknowns are the *generators*. This lattice-generator viewpoint—that the spectrum of a quasi-periodic signal is the integer-lattice image of a few fundamental frequencies, recovered as generators rather than by peak picking—is the organizing principle of Laskar's frequency-map analysis (NAFF) [22, 23] in Hamiltonian and accelerator dynamics; what is specific to Risley identification is that the generators are the *signed* rotation speeds, that the fundamental-line phases and amplitudes are the wedge orientations and magnitudes through the Snell readout of Sec. III, and that generator selection is made a decidable arithmetic problem (Lemma 2, the integer-coverage score) rather than left to resonance inspection. Treating extraction as generator identification (rather than peak picking) is what eliminates, simultaneously, the harmonic-confusion failure mode, the sign enumeration, and the frequency-triple search. The completion stage then inherits a well-posed local problem, consistent with the identifiability analysis.

Our own earlier pipeline (FFT peaks, neural initialization, staged descent; Sec. VII H) reached the same terminal precision *when* its initialization landed in the basin, but required a $56\times 8$-candidate screen and a $9^3$ grid to get there, at $\sim 35$ min per hard case and roughly half reliability in the 9-D protocol (E7). The neural initializer—a physics-informed staged design trained on $10^6$ patterns, not a strawman—is superseded outright: the lattice coordinates make the hard part of the map nearly linear, and nothing remains for a network to learn at $P{=}3$.

### B. Negative results with content

Three predecessor designs of the spectral stage failed during development, and each failure is a reusable lesson now embedded in the theory: residual-guided greedy basis growth is structurally unsound at lattice order $\geq 2$ (Lemma 2); unregularized amplitude solves explode on near-coincident lines while plain ridge *splits* them (hence merge + ridge + sharp-final); and re-extracting fundamentals from refined fits harvests the splitting artifacts (hence extract once, from the selection fit only). We record these because they are exactly the traps a re-implementation would fall into.

Twice, moreover, development-scale ensemble testing (order $10^2$–$10^3$ configurations) falsified the *completeness* of the failure taxonomy—never a theorem, but the implicit claim that every failure had a name—and each time the resolution was a sharper theory: one ensemble exposed the merge-floor term missing from $T_{\text{req}}$ [Eq. (14)], and a single unexplained case in another (healthy amplitudes, clean separations, no TIR; the truth-seeded fit converged to $10^{-25}$) forced condition (v) and the overload retry into Definition 1. We regard this falsification loop—taxonomy, counterexample, derived condition—as part of the method, and the campaign's zero-unexplained target (E1) is its continuation at three orders of magnitude larger scale.

### C. Limitations

The certificates make the limitations quantitative rather than anecdotal: pair separations below $\sim 3\sigma(T)$, wedge angles below the stated detectability threshold, exact lattice relations, and TIR-dominated traces are unrecoverable *at the given* $T$—each with a computable $T_{\text{req}}$, validated at scale by E4. Beyond these: the per-axis reduced Snell model is exact only for the self-consistent inversion (against real hardware the $\mathcal{O}(\alpha_x\alpha_y)$ cross-coupling would enter as model mismatch); the completion stage is implemented for $P{=}3$, its $P$-generalization being mechanical (the readout laws are per-prism) but not yet executed; certificate prism indices refer to the fitted basis and can misalign when that basis is wrong (the run-time feasible-yet-failed self-report remains the detector); cure-time composition for multi-deficit configurations is not derived—only the single-deficit scalings of Sec. V are; global (as opposed to local) uniqueness of the non-paraxial inverse remains open; the campaign is specified and pre-registered but its data are pending (every dependent value is boxed); and this is a simulation study—experimental validation on physical hardware, where the certificate framework should absorb measurement noise as in E6, is the natural next step.

## IX. CONCLUSION

The multi-prism Risley inverse problem is solved by recognizing its algebraic skeleton: the scan pattern lives on the frequency lattice of the signed speeds, the wedge angles are the phases and amplitudes of the fundamental lines, and the remaining parameters are pinned by one well-initialized trust-region solve on the exact model. The resulting inversion recovers the complete $4P+6$-parameter specification from a single pattern with no enumeration and no training, to near machine precision, in seconds—and, we believe more importantly, it *knows what it knows*: every output is accompanied by an error bound whose calibration is testable, or by a quantitative statement of why recovery is impossible at the given observation time, with a prescription for how much observation would suffice. The scaling theory of Sec. VI extends the guarantees to arbitrary prism count—polynomial cost, $\Theta(P^2)$ crowding demand, degree-$K$ relation demand, and a feasibility frontier—and the pre-registered campaign of Sec. VII subjects every statistical claim, including the frontier exponents and the zero-unexplained-failure target, to tests at the $10^6$-configuration scale.

*a. Reproducibility.* All code (the `risley_lattice` package), the campaign harness and every assumption test (`experiments/sweep.py`, `aggregate.py`, A1–A8 scripts), and the raw result records are available at `https://github.com/jbabcanec/Risley_Prism`; random seeds are fixed and case generation is counter-based, so every number in the campaign is reproducible from the integer index range alone.


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